

Optimizing Long-Term User Retention via Multi-Intervention Markov Decision Processes

Sooyeon Park
Ewha Womans University
soopark723@gmail.com

Abstract—This work investigates the impact of interventions by app creators on user retention modeled in a Markov Decision Process-structured environment. Application makers nowadays use recommendation systems and various algorithms to hold onto current users and foster app engagement. Reinforcement learning has been used as a framework to model these systems because of its capacity to model sequential interactions and delayed rewards [5]. Despite the developments of complex algorithms to boost user usage through a particular algorithm, there are still questions surrounding the impacts of various actions that app makers can take and how they interact with one another. In this study, we use an MDP model to explore how interventions including push notifications, added features, added incentives, a combination of notifications and features, and a combination of notifications and incentives, or no intervention, can impact optimal policy for app creators in user retention under a reinforcement learning framework. We discover that a conservative optimal policy prioritizing inaction, with stronger intervention depending on low engagement or high fatigue levels, is most effective in user retainment.

I. INTRODUCTION

Currently, there are many studies on maximizing user retention for applications [2]. Some focus on recommendation systems and their relevance to user life with a deep reinforcement learning framework [1], while others explore how push notifications impact user engagement [4]. However, most focus on just one particular aspect at a time, zeroing in specifically on push notification timing, recommender systems, or user preferences. Unlike previous works focusing on a single intervention type, this study considers a unified intervention space for three intervention types and analyzes how different intervention types interact under fatigue dynamics.

In this experiment, we seek to investigate the effects of notifications, modified features, and in-app incentives in a unified space on long-term user retention. Here, notifications refer to the sending of push notifications to users, modified features refer to the addition or enhancement of in-app tools like a new forum or messaging space, and incentives refer to simple rewards for users such as streaks or discounts. The fatigue dynamics considered for this study correspond to user burnout or loss of engagement, and is directly tied to app intervention actions. The agent – app creators – respond to an environment where diverse factors contribute to user retention and app usage. We use a Markov Decision Process model to frame how these variables interact with the objective of exploring the impacts of diverse app exposures.

II. MARKOV DECISION PROCESS MODEL

The relevant MDP model is defined as follows. It consists of five variables: decision epochs, state space, action space, reward function, and transition probabilities, in the context of an app creator as the agent and the application ecosystem as the environment [5].

A. Decision Epochs

The decision epoch was defined with a finite horizon of discrete time units:

$$t = \{0, 1, 2, \dots, T\},$$

where $T = 50$ and one step corresponds to one day. The time frame was limited to approximately one to two months to capture realistic user churn rates, as many users leave platforms in early usage stages.

B. State Space

The state space was defined as:

$$s_t = (e_t, f_t, u_t, c_t),$$

where,

$$e_t \in [0, 1]$$

represents engagement,

$$f_t \in [0, 1]$$

represents fatigue,

$$u_t \in \{\text{active, moderately engaged, declining}\}$$

represents user stage, and,

$$c_t \in \{0, 1, 2, 3, 4, 5\}$$

represents the churn counter.

C. Action Space

The action space was defined as:

$$a_t \in \begin{cases} 0 : \text{none,} \\ 1 : \text{notify,} \\ 2 : \text{feature,} \\ 3 : \text{incentive,} \\ 4 : \text{notify+feature,} \\ 5 : \text{notify+incentive} \end{cases}$$

The parameters for the action space were defined in terms of engagement boost (α_a), fatigue increment (δ_a), and intervention cost $c(a)$.

D. Reward Function

The reward function was defined as:

$$r_t = e_{t+1} - \lambda f_{t+1} - c(a_t) - \kappa \mathbf{1}_{\text{churn}},$$

where $\kappa = 7$. This is a specific instantiation of the general reward term $r_t = R(s_t, a_t, s_{t+1})$.

The term e_{t+1} represents engagement reward. It represents active users, user retention, and app usage. Higher engagement naturally means a larger reward value. Generally speaking, an engagement value of 0.9 and up could signify “very good” reward contribution, an engagement value of approximately 0.5 could mean “moderate” reward contribution, and a value of 0.1 or less could signify “poor” reward contribution.

The term $-\lambda f_{t+1}$ represents a fatigue penalty, where λ is the discount or decay factor. This term punishes for user annoyance, intervention overuse, and user burnout. The fatigue penalty allows the model to think for long-term strategies. Without it, engagement would always increase, and the oversimplified solution would be to spam incentives forever.

The term $-c(a_t)$ represents intervention cost. It is a cost function to punish taking an action. It takes into account how interventions could have various costs, including: money, engineering effort, downgraded UX, or reduced trust.

The term $-\kappa \mathbf{1}_{\text{churn}}$ represents churn cost. Setting kappa to a large value like 7 indicates that churn is extremely undesirable. The agent becomes willing to take action to avoid churn, which could come in the form of spending on expensive incentives to rescue declining users. This operates under the pretense that app makers prefer to spend resources over losing users entirely.

Together, these terms allow the reward function to account for delayed trade-offs, sparse interventions, and strategic intervention timing.

E. Transition Probabilities

The transition probabilities were factorized as:

$$\begin{aligned} P(e_{t+1}, f_{t+1}, u_{t+1}, c_{t+1} \mid s_t, a_t) &= P(e_{t+1} \mid s_t, a_t) \\ &\times P(f_{t+1} \mid s_t, a_t) \\ &\times P(u_{t+1} \mid s_t) \\ &\times P(c_{t+1} \mid s_t). \end{aligned} \quad (1)$$

Although this factorization assumes conditional independence among the components of the transition distribution, engagement and fatigue remain coupled in the underlying dynamics because fatigue directly influences the engagement transition mean. Consequently, the independence assumption is adopted solely for tractability and applies only to the probabilistic factorization rather than the deterministic state-update equations.

1) *Engagement Transition Probability*: The engagement transition probability was modeled as a Gaussian random variable,

$$e_{t+1} \sim \mathcal{N}(\mu_t, \sigma^2),$$

where $P(e_{t+1} \mid s_t, a_t)$ induced by this distribution. The transition mean was defined as:

$$\mu_t = e_t + \alpha_a(1 - 0.6f_t) - \beta f_t^{1.05} - \eta(e_t - \bar{e}_u).$$

Values were clipped to the interval $[0, 1]$ after sampling.

The complete transition can be written as,

$$e_{t+1} = \mu_t + \varepsilon_t,$$

where $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$.

The parameter α_a refers to action effect, capturing how interventions (notifications, features, incentives) influence engagement. The term $-\beta f_t^{1.05}$ refers to fatigue penalty, meaning that tired users engage less. There are diminishing returns for more actions because too many interventions can lead to user burnout. The term $-\eta(e_t - \bar{e}_u)$ refers to mean reversion, meaning that users return to their “natural level” $\bar{e}_u$. Users with high engagement cool down, and users with low engagement come back up. Finally, ε_t captures randomness, meaning there is a degree of unpredictability in user behavior that could come from mood, external factors, and general arbitrariness.

2) *Fatigue Transition Probability*: The fatigue transition probability was defined as:

$$P(f_{t+1} \mid s_t, a_t) = \begin{cases} 1, & f_{t+1} = \min(1, \gamma f_t + \delta_a), \\ 0, & \text{otherwise.} \end{cases}$$

The fatigue dynamics follow $f_{t+1} = \min(1, \gamma f_t + \delta_a)$, which produces a deterministic transition. Values were clipped to the interval $[0, 1]$.

The parameter γ controls fatigue decay, which means fatigue fades over time, as less interventions could lead to recovered engagement. The parameter δ_a captures fatigue accumulation from actions, meaning that every action increases fatigue, with notifications and incentives still causing user withdrawal.

3) *User Stage Transition Probability*: User stage transitions were defined as:

$$u_{t+1} = \begin{cases} \text{active,} & e_{t+1} > 0.6, \\ \text{moderately engaged,} & 0.3 < e_{t+1} \leq 0.6, \\ \text{declining,} & e_{t+1} \leq 0.3. \end{cases}$$

4) *Churn Transition Probability*: The churn counter transition was defined as:

$$c_{t+1} = \begin{cases} c_t + 1, & e_{t+1} < 0.25, \\ 0, & \text{otherwise.} \end{cases}$$

The churn transition is defined with churn state s_{churn} . If engagement remains below the threshold for five consecutive steps, corresponding to $c_t \geq 5$, then $P(s_{t+1} = s_{\text{churn}} \mid s_t, a_t) = 1$. Once entered, the churn state becomes absorbing:

$P(s_{t+1} = s_{\text{churn}} \mid s_t = s_{\text{churn}}) = 1$. No additional rewards are received after churn, and the user is removed from the simulation for all remaining decision epochs. This design reflects the common conception that prolonged disengagement is a greater indicator of churn risk than a single low-engagement event [2].

III. METHODOLOGY: DQN LEARNING

This project used an objective function, the Bellman equation, and Q-learning on simulated data to approximate the optimal policy for the previously defined Markov Decision Process model. We first designed simulated data to show how different app exposures can impact user engagement. The simulated data included 13 categories with the following information: user ID, time step (t), current engagement, current fatigue, user type (active/moderately engaged/declining), current churn count, chosen action, received reward, next engagement (for time step $t+1$), next fatigue (for time step $t+1$), next user type (change in active/moderately engaged/declining state), updated churn count (beginning churn count for step $t+1$), and churn status (0 or 1). Sample simulated data can be seen in Appendix A.

We created data points for 5,000 users (episodes) across 50 decision epochs, resulting in approximately 250,000 observations excluding episodes terminated by churn.

The overall objective is,

$$\max_{\pi} \mathbb{E} \left[\sum_{t=0}^T \gamma^t r_t \right], \quad (2)$$

where r_t denotes the reward at time t , γ is the discount factor, and π is the policy.

The Bellman optimality equation is,

$$V(s) = \max_a [r(s, a) + \gamma \mathbb{E}_{s'} [V(s')]]. \quad (3)$$

It recursively measures the value of a state by considering the immediate reward and the optimal future value. Actions were selected according to an ϵ -greedy exploration policy. With probability $1 - \epsilon$, the agent selected the action with the highest estimated value, and with probability ϵ , a random action was chosen [5].

Deep Q-Learning (DQN) was implemented to approximate action-value functions using a neural network [6]. It was chosen over traditional tabular Q-learning because engagement and fatigue were modeled as continuous variables, which would create an unmanageably large state space. DQN allowed for approximated values and thus generalization across unseen states, preferable to saving individual discrete state-action pairs. The Q-function is the action-level value, which shows the value of taking action a in state s .

The action-value function is defined as:

$$Q(s, a) = r(s, a) + \gamma \mathbb{E}_{s'} \left[\max_{a'} Q(s', a') \right]. \quad (4)$$

The Q-learning update is the iterative learning rule that allows you to obtain your new estimate by combining the old estimate

with an error variable (derived from subtracting expected outcome from observed outcome). It is defined as:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right], \quad (5)$$

where α is the learning rate.

Each user state was represented as a continuous feature vector consisting of engagement, fatigue, normalized churn risk, and one-hot encoded user-stage indicators. Transition tuples (s_t, a_t, r_t, s_{t+1}) were stored in an experience replay buffer and sampled to train the neural network approximation of the Q-function with stochastic gradient updates.

The learned policy was extracted as:

$$\pi(s) = \arg \max_a Q(s, a). \quad (6)$$

Experience replay, coupled with the use of a separate target network, allowed for more stabilized learning, with the former randomly sampling past transitions to lessen time correlations between observations, and the latter periodically froze target Q-values during optimization to stabilize updates [6]. The pseudocode for the Q-learning algorithm can be found in Appendix B.

We initially hypothesized that the expected optimal policy can be grouped by the presence of fatigue. The absence of fatigue would mean it would likely be optimal to always put in an intervention. The presence of fatigue would mean it would likely be optimal to vary action choice depending on engagement levels: low engagement $\rightarrow$ incentive, notify + incentive, notify + feature; low engagement $\rightarrow$ feature; medium to high engagement $\rightarrow$ do nothing; high fatigue $\rightarrow$ no action. In the absence of fatigue penalties, the reward simplifies to $r_t = e_t$. Long-term disadvantages of repeated actions decrease, so the agent was expected to become more intervention-heavy. Action costs and churn dynamics were still expected to, however, prevent overly aggressive intervention policies. On the other hand, when fatigue penalties are included, the reward becomes: $r_t = e_{t+1} - \lambda f_{t+1} - c(a_t) - \kappa \mathbf{1}_{\text{churn}}$. We expected that there would be a tradeoff where taking an action led to more engagement, but also to more fatigue, which would create a future penalty. The Bellman target, $r + \gamma \max_{a'} Q(s', a')$, captures the tradeoff between immediate engagement gains and future fatigue costs. Consequently, a sparse policy was expected to emerge. Frequent actions would lead to high fatigue, and high fatigue would lead to lower future value. So, it was expected that app makers should take a cautious approach in which they took an action only when the marginal benefit was greater than the future cost.

IV. EXPERIMENTS

The basic settings for the experiment were as follows. A total of 5,000 episodes, or trajectories of user data, were simulated, each spanning 50 days. Target and policy networks were created to track trajectories and Q-values. The discount factor was set to 0.95, the learning rate to 0.001, batch size to 64, the replay buffer size to 50,000, and the target network update frequency to 50. Thus, the replay memory

buffer could hold up to 50,000 transition experiences, and random minibatches of size 64 were sampled during training. The target network was synchronized with the policy network every 50 episodes.

The dynamics parameters were set as follows: fatigue penalty coefficient $\beta = 0.36$, mean reversion coefficient $\eta = 0.15$, noise standard deviation $\sigma = 0.07$, and fatigue decay coefficient $\gamma = 0.68$. The reward parameters were fatigue weight $\lambda = 0.32$ and churn penalty $\kappa = 7.0$.

The action space was defined from 0 through 5 with six possible interventions: none, notify, feature, incentive, notify+feature, and notify+incentive. Each action was characterized by an engagement boost α_a , a fatigue increment δ_a , and an intervention cost $c(a)$. The parameter assignments are shown in Table I.

TABLE I: Action parameter designations. Specific parameter assignments for engagement boost, fatigue, and cost are seen here, according to each of six possible action interventions.

Action	Meaning	Engagement Boost (α_a)	Fatigue (δ_a)	Cost ($c(a)$)
0	None	0.00	0.00	0.00
1	Notify	0.04	0.08	0.03
2	Feature	0.07	0.08	0.05
3	Incentive	0.11	0.15	0.10
4	Notify + Feature	0.10	0.18	0.14
5	Notify + Incentive	0.15	0.26	0.20

The baseline engagement levels associated with user stages were defined as 0.7 for *active*, 0.5 for *moderately engaged*, and 0.4 for *declining* users.

Initial engagement values were sampled from the interval $[0.3, 0.8]$, while initial fatigue values were sampled from $[0.0, 0.2]$. Users were classified according to:

$$u_t = \begin{cases} \text{active,} & e_t > 0.6, \\ \text{moderately engaged,} & 0.3 < e_t \leq 0.6, \\ \text{declining,} & e_t \leq 0.3. \end{cases}$$

The churn counter was initialized to 0. Additionally, churn resulted in immediate episode termination, meaning that no further rewards or state transitions were generated for churned users.

The model followed an ϵ -greedy exploration policy in which the exploration probability decayed exponentially from 0.20 to a minimum value of 0.02 during training. This provided a gradual transition from exploration to exploitation and helped prevent premature convergence to suboptimal policies.

To model diminishing intervention effectiveness, the engagement contribution of each action was multiplied by the fatigue multiplier $(1 - 0.6f_t)$. Consequently, higher fatigue reduced the effectiveness of interventions in generating engagement gains.

A nonlinear fatigue penalty was also incorporated to model slightly nonlinear burnout effects. Rather than using the linear term $-\beta f_t$, the fatigue penalty was specified as $-\beta f_t^{1.05}$. This played the role of modeling the negative impacts of chronic fatigue that would lead to faster disengagement, incorporating the idea that long-term, constant intervention could lead to more user burnout.

The DQN architecture was made up of a fully connected feedforward neural network with two hidden layers. Each layer contained 64 neurons and used ReLU activation. User stage categories – active, moderately engaged, declining – were one-hot encoded before being put into the network. Thus, the input layer took in as input a six-dimensional encoded state vector consisting of: engagement, fatigue, normalized churn risk, and the three one-hot encoded user stage indicators. The output layer generated a vector with six approximations of Q-values representing each possible intervention action. The model used the Adam optimizer with a learning rate of 0.001 for network optimization, and mean squared error (MSE) to find the distance between Q-values and Bellman target estimates. Training was executed with online interaction with the environment. The output experience tuple following every state transition was added to the replay buffer, to be used for iterative gradient updates.

Additionally, four ablation studies were conducted. In Ablation Study 1, the exploration parameter was fixed at $\epsilon = 0$ to evaluate the effects of removing exploration. Ablation Studies 2 and 3 modified the churn penalty parameter κ , reducing it to 3.0 and increasing it to 12.0, respectively. Ablation Study 4 removed the churn counter from the state representation entirely. These studies were designed to examine how exploration, churn penalty severity, and churn-state observability influence policy quality and retention outcomes within the simulated environment.

V. RESULTS

A. Optimal Policy Interpretation

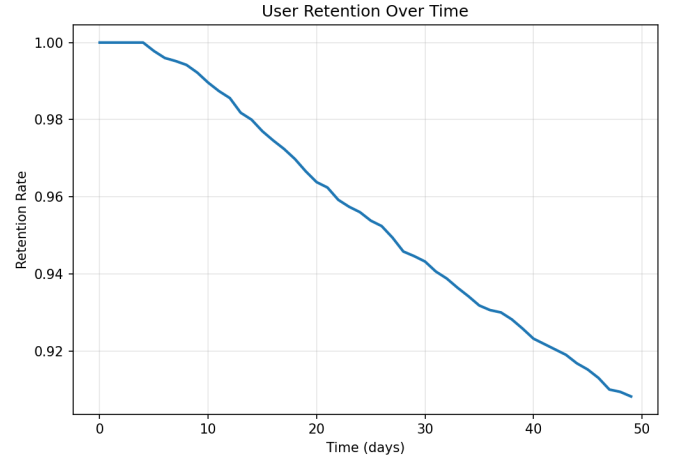


Fig. 1: User retention curve. User retention shows a steadily declining trend over 50 days.

Results for changes in user engagement, fatigue, retention, and the resulting optimal policy point to a consistent learning environment and learned strategy.

As shown in Figure 1, retention began at 100% on Day 1 and declined to approximately 91% by Day 50. There were 468 churn events out of 5,000 total episodes, corresponding

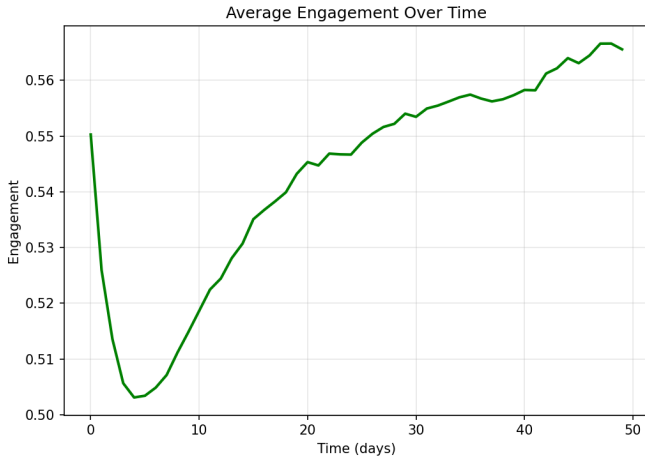


Fig. 2: Average engagement curve. User engagement initially declines, then maintains a steadily increasing trend over 50 days.

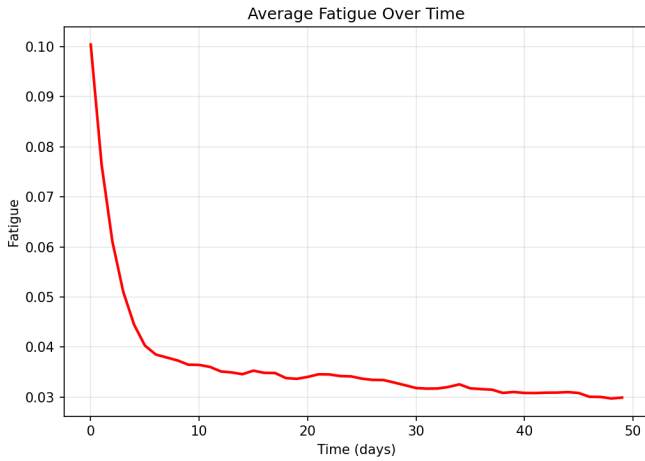


Fig. 3: Average fatigue curve. User fatigue reflects a declining trend over 50 days.

to a churn rate of 9.36% and an overall retention rate of approximately 90.64%. This relatively high retention rate suggests that the learned policy was effective at maintaining long-term user engagement.

Consistent with the retention results were the engagement and fatigue trends shown in Figures 2 and 3. Average engagement began at approximately 0.55, declined to roughly 0.50 after four days, and then increased steadily to approximately 0.566 by Day 50. Fatigue followed the opposite pattern, beginning near 0.10 on Day 1, decreasing sharply to approximately 0.04 after four days, and then gradually declining to approximately 0.03 by Day 50. These trends indicate that the policy successfully learned to strategize for long-term impacts to show favorable outcomes for both engagement (an increase) and fatigue (a decrease).

Results for learned policy are observable in Table 2 and Figure 4. The dominant action was *none*, accounting for 89.76%

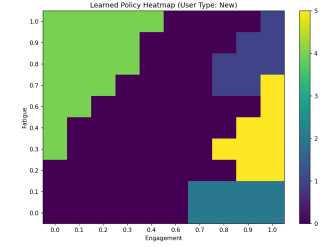


Fig. 4: Learned policy heatmap. The heatmap shows what intervention actions were taken in terms of differing degrees of fatigue and engagement rates.

TABLE II: Policy action distribution. The learned policy distributes actions according to the six possible interventions, with no action being the highest, and “notify+feature” being the lowest.

Action	Percentage
0: None	89.76%
1: Notify	0.78%
2: Feature	6.93%
3: Incentive	1.19%
4: Notify + Feature	0.48%
5: Notify + Incentive	0.85%

of decisions. For instances of low to mid-engagement coupled with low to mid-fatigue, the learned policy was no action. This is aligned with the concept of a taking on a conservative learned policy, that aims to strategize for intervening only when necessary in consideration of long-term impacts. Low engagement combined with moderate-to-high fatigue often triggered the “notify+feature” intervention, despite its relatively low overall frequency (0.48%). High engagement and moderate fatigue occasionally resulted in “notify+incentive” interventions (0.85%), while high engagement and low fatigue most commonly triggered the “feature” intervention (6.93%). High engagement and high fatigue generally led to the lower-cost “notify” intervention (0.78%), reflecting the preference to avoid stronger interventions when users already exhibit signs of fatigue. Overall, very low engagement states tended to receive stronger interventions, whereas high-engagement states exhibited more nuanced behavior depending on fatigue levels. High engagement combined with high fatigue led to relatively mild interventions such as notifications, while high engagement combined with low fatigue allowed more aggressive actions such as features or notify-plus-incentive combinations.

TABLE III: A summary of policy actions in accordance with varying engagement and fatigue levels.

Engagement	Fatigue	Action
Low	Low	None
Low	Moderate	Notify + Feature
Low	High	Notify + Feature
Medium	Low	None
Medium	Moderate	None
Medium	High	None
High	Low	Feature
High	Moderate	Notify + Incentive
High	High	Notify

TABLE IV: Summary of ablation studies. Action distribution, policy results, and retention for each of five cases can be seen here, with the baseline “original” study at the leftmost column.

Metric	Original	Ablation 1 ($\epsilon \approx 0$)	Ablation 2 ($\kappa = 3$)	Ablation 3 ($\kappa = 12$)	Ablation 4 (No Churn State)
Churn rate	9.36%	24.50%	8.72%	10.44%	13.76%
Avg trajectory length	47.80	43.95	47.73	47.43	46.56
Avg reward	0.509	0.446	0.522	0.493	0.509
% none action	89.76%	79.73%	90.66%	88.73%	89.50%
% notify	0.78%	2.43%	0.86%	0.91%	0.92%
% feature	6.93%	12.30%	6.19%	7.91%	7.65%
% incentive	1.19%	1.93%	1.09%	0.95%	1.04%
% notify + feature	0.48%	1.19%	0.47%	0.47%	0.49%
% notify + incentive	0.85%	2.42%	0.73%	1.03%	0.41%
Total transitions	239,008	43,945	238,626	237,144	232,785
Episodes	5,000	1,000	5,000	5,000	5,000
Policy diversity	Moderate	High	Moderate	Moderate	High
Learned behavior	Conservative, fatigue-aware	Over-intervention	Conservative, balanced	Overly cautious	Reactive, short-term
Retention quality	Strong	Weak	Strongest	Moderate	Weak-Moderate
Stability	High	Low	High	Moderate	Moderate

The learned policy adopted a strategy that had a hesitant approach. The agent found it best to aim for low fatigue and minimized intervention costs to garner best long-term value, as opposed to frequent engagement attempts and aggressive interventions for low-engagement users. The resulting optimal policy reserved strong interventions for areas with very low engagement or low fatigue, and relied mainly on inaction as the strategy.

B. Ablation Studies

Table IV summarizes the results of the four ablation studies.

For Ablation Study 1, where ϵ was set to 0 and no exploration was implemented, the churn rate increased to 24.50%, more than doubling the baseline churn rate of 9.36%. These results demonstrate the necessity of exploration. Allowing exploitation to dominate resulted in shorter episode trajectories and lower rewards. Premature convergence led the agent to learn a suboptimal policy, ultimately producing substantially higher churn rates.

In Ablation Studies 2 and 3, the churn penalty parameter κ was modified. When κ was reduced from 7 to 3, the churn rate decreased slightly from 9.36% to 8.72%. Conversely, increasing κ to 12 raised the churn rate to 10.44%. These findings suggest that a lower churn penalty allowed the agent to learn a policy requiring fewer interventions, reducing fatigue accumulation and ultimately improving retention. In contrast, an excessively large churn penalty encouraged overly aggressive intervention behavior, increasing long-term fatigue and resulting in higher churn rates.

In Ablation Study 4, the churn counter was removed from the state representation. Consequently, the DQN could no longer distinguish between users who had recently become disengaged and users who had remained disengaged for multiple consecutive days. The agent only observed engagement, fatigue, and user stage at each decision epoch. Without access to the churn counter, intervention timing became less precise, resulting in shorter trajectories and increased churn. The churn rate rose to 13.76%, representing approximately a 47% increase relative to the baseline.

Higher churn rates were generally associated with more aggressive intervention policies. For example, in Ablation Study 1, where churn more than doubled, the proportion of “none” actions decreased from 89.76% to 79.73%, while stronger interventions increased substantially. The frequency of the “feature” action increased to 12.30%, while “notify” and “notify+incentive” increased to 2.43% and 2.42%, respectively.

The ablation studies showed how exploration and modifying churn penalty in a controlled environment impacted outcomes. Exploration was necessary to prevent premature convergence to a suboptimal policy that could lead to higher churn rates. Churn penalty in varied degrees had varied results: too little churn penalty leads to under-intervention, too much churn penalty leads to over-intervention and fatigue, and moderate churn penalty shows to have the best balance. A combination of the presence of limited exploration, and a moderate churn penalty showed to have the most success in gaining user retention. Specific engagement, fatigue, user retention, and policy trends for each ablation study can be found in Appendix C.

VI. CONCLUSION

In this study, we explored the impacts of different interventions by app makers and their effect on user retention in a reinforcement learning framework, and found that an optimal policy prioritizing inaction, coupled with stronger interventions dependent on low engagement or high fatigue, was most effective in long-term user retention. We structured the problem as a Markov Decision Process and implemented a Deep Q-Network on simulated user trajectories to model how actions such as notifications, features, or incentives impacted user churn and retention over a limited time frame.

The learned policy favored a cautious approach over an aggressive one, avoiding excessive intervention policies such as “notify+incentive” or “notify+feature.” It instead took to non-action for most of the time, and this proved to be effective in maintaining lower fatigue values while showing steadily inclining engagement over the long term. Stronger actions were only implemented in cases of very low user

engagement. The agent learned to strategize for long-term by taking into account fatigue and churn penalties. The optimal policy successfully achieved high user retention rates, while also lowering fatigue and boosting user engagement.

Four ablation studies were carried out to explore the effects of exploration and churn-related dynamics in the controlled simulated environment. Exploration turned out to be necessary for an optimal learned policy, as exploiting led to premature convergence policies which doubled churn rates. Adjusting churn penalty revealed that a harsher penalty could lead to over-intervention and a more lenient penalty could lead to under-intervention, showing a balanced, moderate churn penalty to be the most effective in contributing to an optimal policy.

The limitations of this paper are as follows. The environment and user trajectories implemented were simulated data, and the interacting variables of action interventions were simplified for the sake of the experiment. Real user data and app usage dynamics are far more complex and diversified. They take into consideration demographic, gender, and social factors, as well as incorporating user preferences and diverse reasons for user burnout. This study could be further improved by implementing real-world data with accurate values to measure user background, app usage, and app dynamics.

This study ultimately reveals that maximizing user retention relies more on a balancing strategy that considers long-term user fatigue and churn, over maximizing the number of timed interventions. Implementing the reinforcement learning framework is an effective guide to creating a balanced framework for adaptive user strategy in app creation and user retention.

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APPENDIX

TABLE V: Example trajectory from a single simulated episode.

Episode	t	e_t	f_t	User Type _{t}	c_t	Action	Reward	e_{t+1}	f_{t+1}	User Type _{$t+1$}	c_{t+1}	Churn
0	0	0.7543	0.0335	active	0	none	0.7360	0.7433	0.0228	active	0	0
0	1	0.7433	0.0228	active	0	none	0.6675	0.6725	0.0155	active	0	0
0	2	0.6725	0.0155	active	0	none	0.7512	0.7546	0.0105	active	0	0
0	3	0.7546	0.0105	active	0	none	0.6281	0.6304	0.0072	active	0	0
0	4	0.6304	0.0072	active	0	none	0.5855	0.5871	0.0049	moderately engaged	0	0
0	5	0.5871	0.0049	moderately engaged	0	none	0.4278	0.4288	0.0033	moderately engaged	0	0
0	6	0.4288	0.0033	moderately engaged	0	none	0.3607	0.3614	0.0023	moderately engaged	0	0
0	7	0.3614	0.0023	moderately engaged	0	none	0.2573	0.2577	0.0015	declining	0	0
0	8	0.2577	0.0015	declining	0	feature	0.3890	0.4649	0.0810	moderately engaged	0	0
0	9	0.4649	0.0810	moderately engaged	0	none	0.4763	0.4940	0.0551	moderately engaged	0	0

Algorithm 1 Deep Q-Network with Experience Replay for User Retention Optimization

Initialize replay memory D with capacity N

Initialize policy network $Q(s, a; \theta)$

Initialize target network $Q_{\text{target}}(s, a; \theta^-)$

$\theta^- \leftarrow \theta$

for episode = 1, ..., M **do**:

Initialize state $s_1 = (e_1, f_1, u_1, c_1)$

for $t = 1, \dots, T$ **do**:

Encode state $x_t = [e_t, f_t, c_t/5, \text{one-hot}(u_t)]$

Choose action using ϵ -greedy policy

$$a_t = \begin{cases} \text{random action,} & \text{with probability } \epsilon \\ \arg \max_a Q(x_t, a; \theta), & \text{otherwise} \end{cases}$$

Execute action a_t

Observe next state,

$$x_{t+1} = [e_{t+1}, f_{t+1}, c_{t+1}/5, \text{one-hot}(u_{t+1})]$$

Compute reward,

$$r_t = e_{t+1} - \lambda f_{t+1} - c(a_t) - \kappa \mathbf{1}_{\text{churn}}$$

Observe terminal indicator d_t

Store transition $(x_t, a_t, r_t, x_{t+1}, d_t)$ in replay memory D

Sample minibatch $(x_j, a_j, r_j, x_{j+1}, d_j)$ from D

for each sampled transition **do**:

if $d_j = 1$ **then**:

$$y_j = r_j.$$

else:

$$y_j = r_j + \gamma \max_{a'} Q_{\text{target}}(x_{j+1}, a'; \theta^-)$$

end if

end for

Compute loss:

$$L(\theta) = \frac{1}{B} \sum_j \left(y_j - Q(x_j, a_j; \theta) \right)^2$$

Perform gradient descent update

if episode mod $C = 0$ **then**:

$$\theta^- \leftarrow \theta$$

end if

if churn detected **then**:

terminate episode

end if

end for

end for

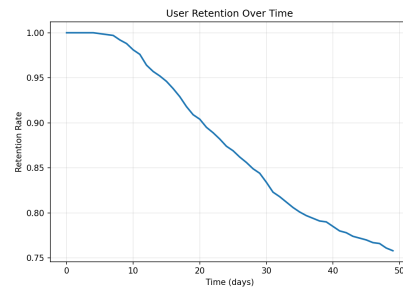


Fig. 5: Retention with no exploration.

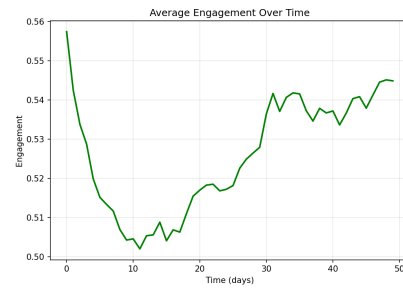


Fig. 6: Engagement with no exploration.

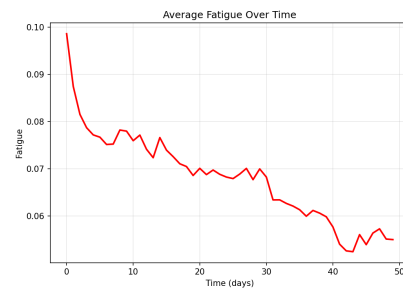


Fig. 7: Fatigue with no exploration.

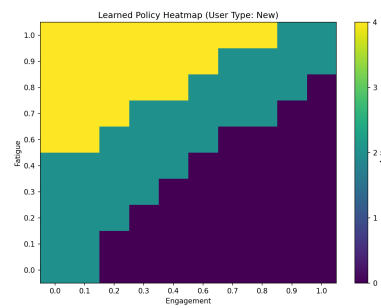


Fig. 8: Policy with no exploration.

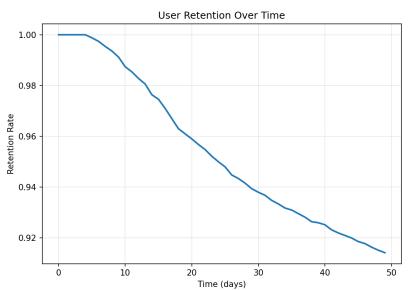


Fig. 9: Retention when $\kappa = 3$.

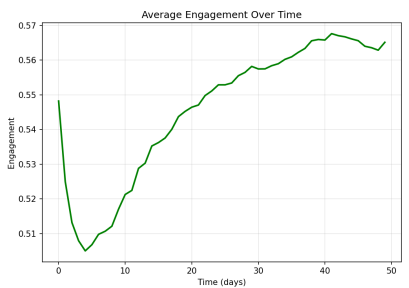


Fig. 10: Engagement when $\kappa = 3$.

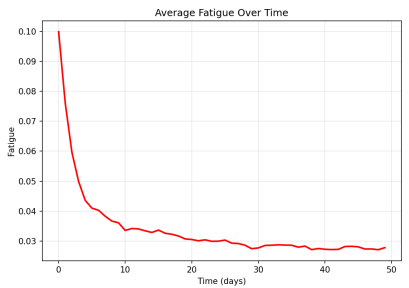


Fig. 11: Fatigue when $\kappa = 3$.

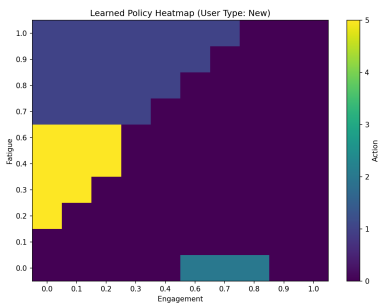


Fig. 12: Policy when $\kappa = 3$.

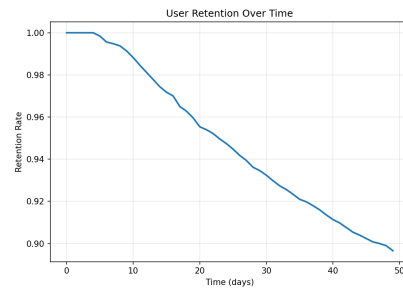


Fig. 13: Retention when $\kappa = 12$.

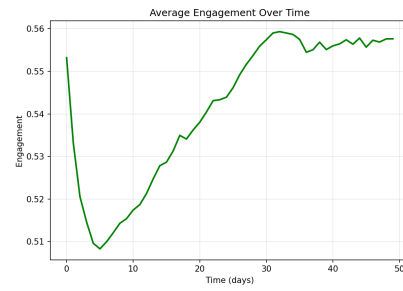


Fig. 14: Engagement when $\kappa = 12$.

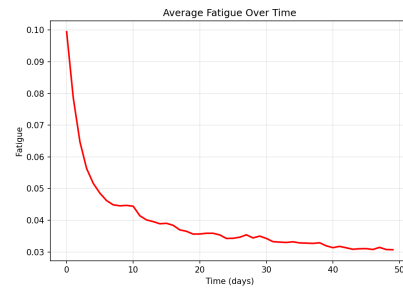


Fig. 15: Fatigue when $\kappa = 12$.

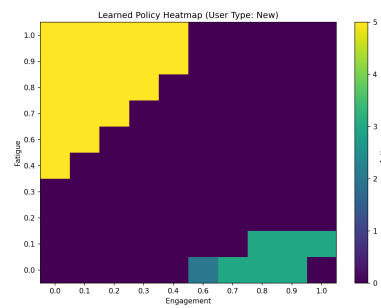


Fig. 16: Policy when $\kappa = 12$.

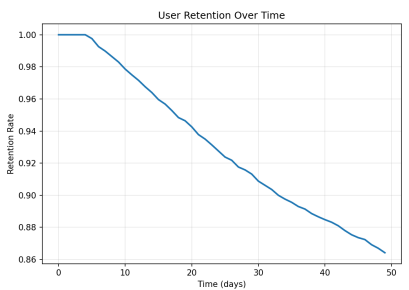


Fig. 17: Retention when churn counter removed.

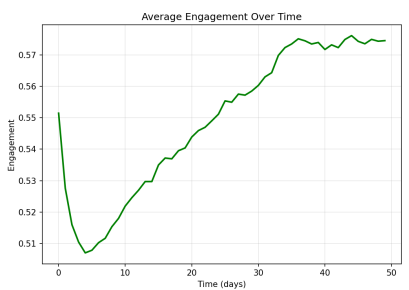


Fig. 18: Engagement when churn counter removed.

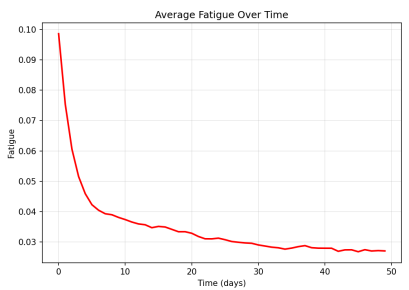


Fig. 19: Fatigue when churn counter removed.

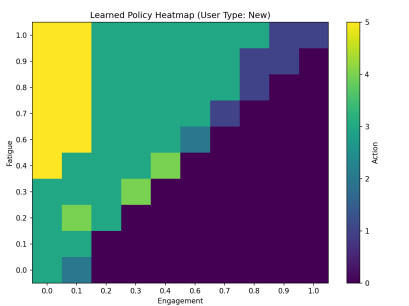


Fig. 20: Policy when churn counter removed.